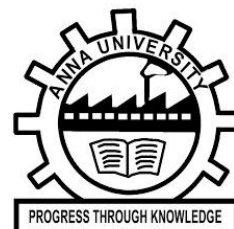


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BTech-Artificial intelligence and Data Science

Anna University Regulation: 2021

AD3391- Database Design and Management

II Year/III Semester

Hand Written Notes

Unit- V

Prepared By,

Mrs.S.Kiruthika, AP/AIDS

Unit - V Object Relational & No SQL Databases

Object Database Concept:

⇒ It is designed for business applications such as inventory, employee, university, bank, library, air-line reservation systems & so on.

⇒ It simple data types that are well suited to the relational data model.

⇒ Object based database. provide the solution to model the real world object & their behaviour.

⇒ It is a relational database model.

Mapping EER to ODB Schema:-

⇒ Object oriented paradigm encapsulate data & corresponding methods in a single entity called object.

⇒ The object-oriented data model is a logical data model just like the ER model.

⇒ It helps in adapting object oriented paradigm to database systems.

Object & Object Identifier:-

In real world entity is modeled as object. It has unique id. Maintained which is called object identifier (OID). Object identifiers used to identify objects. Object identifiers are unique. No two objects have the same identifier. Each object has only one object identifier.

Attribute & Method:-

Every object has state & behavior. The state is a set of values for attributes of the object. The behavior is set of methods. For example, object named circle, has state radius & computing its area is a behavior.

Class:

Class is a means of grouping all objects which share the same set of attributes & methods. An object must belong to only one class as an.

instance of that class

Class Hierarchy & Inheritance:

The class hierarchy is used to represent the inheritance property of object oriented paradigm. The inheritance is a mechanism in which a new class is derived from existing class.

Object structure:-

⇒ Object is a fundamental unit of object oriented programming in which data value & code is encapsulated in a single unit.

⇒ The objects of rest of the system are carried out by using messages. The interface among various objects of rest of the system is messages.

An object has associated with it.

⇒ A set of variables that contain the data for the object.

⇒ A set of messages to which the object responds.

⇒ A set of methods, each of which is a body of code to implement each

message, a method returns a value as the response to the message.

⇒ only variables in the object itself may be referenced directly.

⇒ Data in other objects are referenced only by sending messages.

⇒ Methods can be read only or update methods. Read only methods do not change the value of the object.

eg: The Object circle can have methods such as compute_area()

Object classes:

⇒ Similar objects are grouped into classes. In other words, object is an instance of a class. For example,

class student {

* variables *

int RollNo;

String Name;

String address;


```

String get_name();
String get_address();
int get_Roll NO();

```

2

Method can be defined separately. for
example:-

```

int get_RollNO()
{

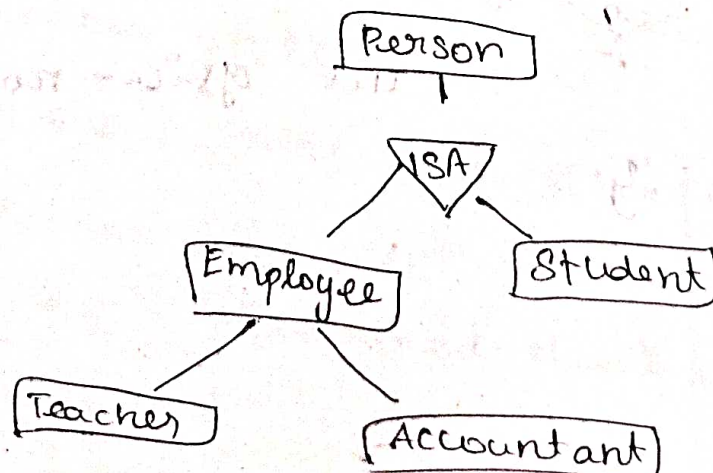
```

Return RollNo;

2

Inheritance:-

=> In object oriented database schema, there can be large number of classes. Some classes are similar and can be derived from old existing class. The classes are placed into specialization or is - a hierarchy.




```

class Person
{
    string name;
    string address;
}

class Student : public Person
{
    int Roll NO;
    string course_taken;
}

class Employee : public Person
{
    data join - data;
    int salary;
}

class Teacher : public Employee
{
    string subject;
}

class Accountant : public Employee
{
    int office-no;
}

```

The keyword `is` is used to indicate that a class is a specialization of another class.

⇒ class hierarchy and inheritance of properties from more general classes.

⇒ The important benefit of using inheritance is code-reusability. This can be achieved by means of substitutability property.

⇒ Substitutability means any method of a class, say Person, can be invoked equally well with any object belonging to any subclass, such as subclass Teacher of Person.

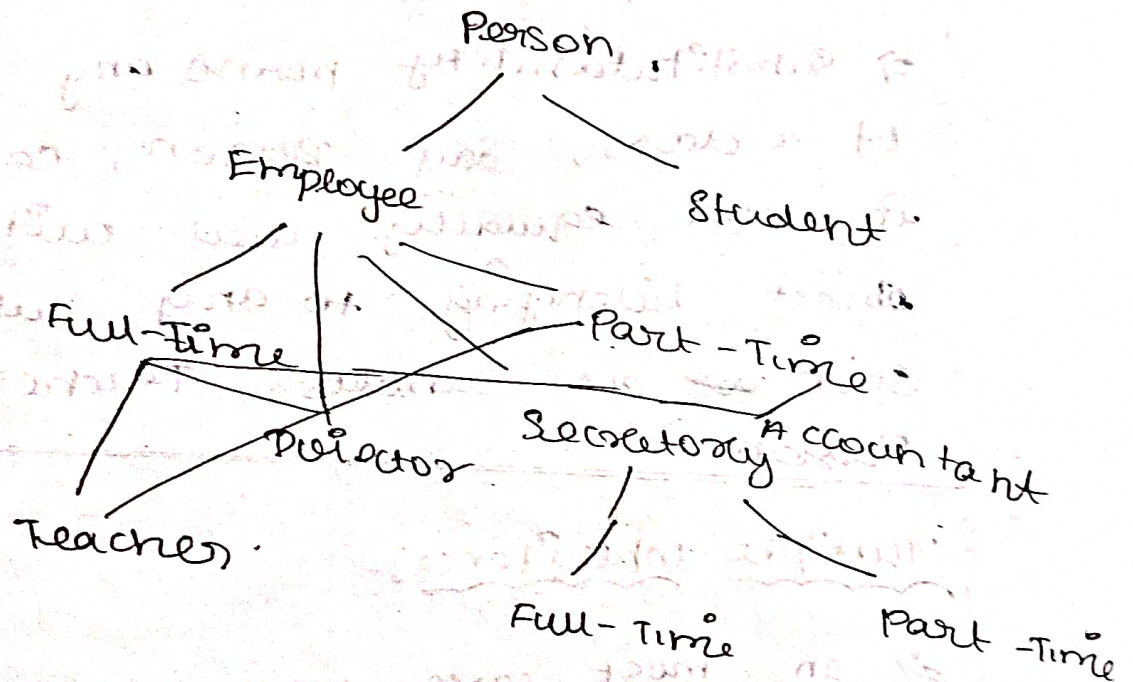
Multiple inheritance:-

⇒ In most cases, tree-structured organization of classes is adequate to describe applications.

⇒ However, there are situations that cannot be represented well in a tree-structured class hierarchy.

⇒ To avoid the conflict between two occurrences, we use multiple inheritance.

⇒ Multiple inheritance is an ability of class to inherit variables from multiple super classes. The class-subclass relationship is represented by directed acyclic graph (DAG).



Handling name conflicts: When multiple inheritance is used, there is potential ambiguity if the same variable or method can be inherited from more than one super class.

Object Identity

⇒ For identifying the RDBMS record uniquely in the table, we use primary key associated with it. In object oriented database management system we use Object ID to identify the record.

⇒ Associated with each object there is a unique ID maintained which is called Object Identifier (OID).

⇒ Object identity is typically implemented via a unique, system-generated OID. The value of the OID is not visible to the external user, but is used internally by the system to identify each object uniquely and to create and manage inter-object references.

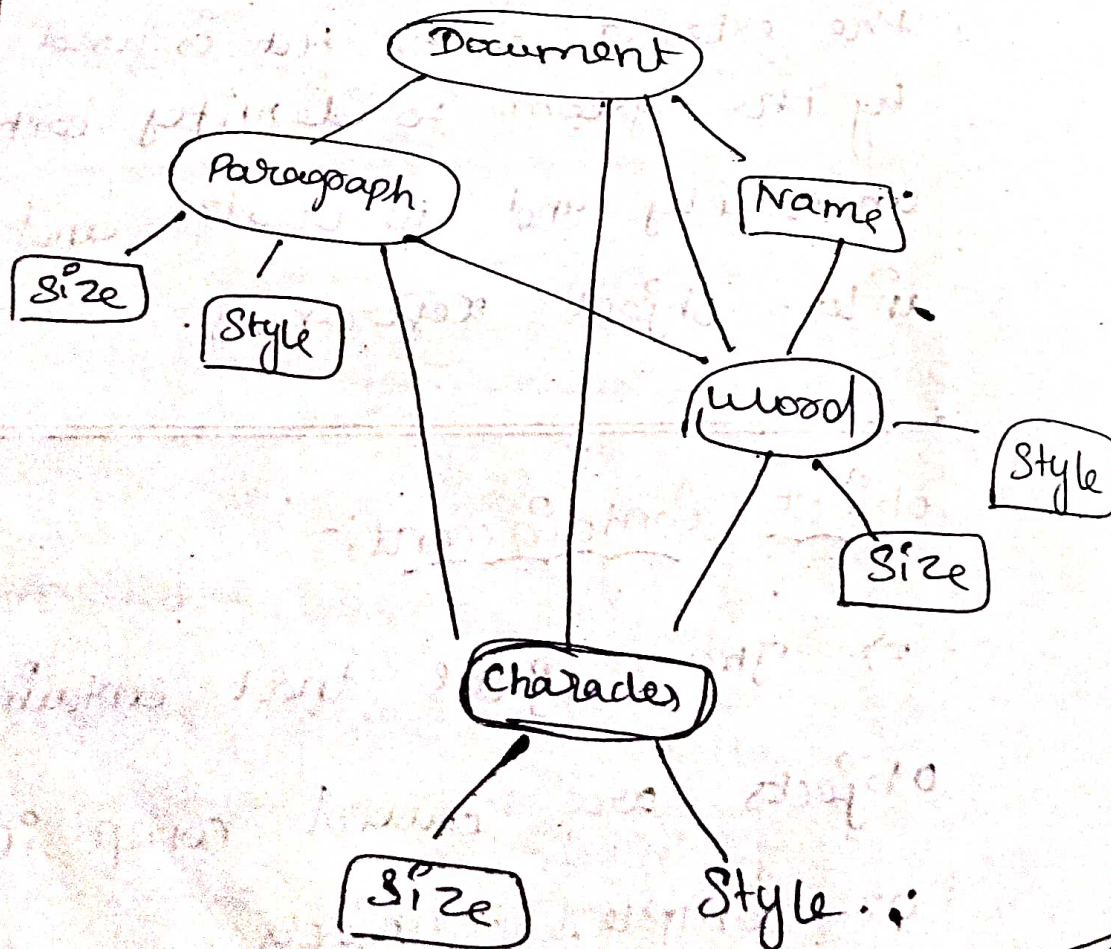
Object Containment

⇒ The objects that contain another objects are called composite objects or complex objects.

⇒ There can be multiple levels of objects explanation. These levels can be represented using containment hierarchy.

⇒ The links between classes must be interpreted as is-part-of rather than the is-a interpretation of links in an inheritance hierarchy.

⇒ The containment hierarchy can be used to find all object contained in document object.



Reference types:-

⇒ Reference types are all types other than value types. Variables of reference types store a reference to the memory address of the value.

⇒ A reference type value can be stored in one table and used as a reference to specific row in some other table.

⇒ To use a reference type in an SQL statement, use REF, where type-name represents the referenced type.

⇒ REF is SYSTEM GENERATED in a CREATE TYPE statement indicates that the actual values of associated REF type are provided by the system.

Row Types:-

⇒ A row type is a sequence of name-data type pair. It is used to provide a data type to the rows in the table.

⇒ Using row type complete row can be stored in a variable. This variable can be passed as an argument to

the routine or we can get the returned value from a function call within this variable.

consider an Employee table containing Emp-Id and address and insert into the new table.

```
CREATE TABLE Employee(
  empID CHAR(5)
  address Row (Street VARCHAR(30),
  City VARCHAR(25),
  PinCode Row (City-Id VARCHAR(6),
  Region VARCHAR(10)));
INSERT INTO Employee
VALUES ('Account-1234' Row ('Shankar',
  'Chennai', Row ('Chennai', 'And Nagar'));
```

User Defined Types - Routines:-

⇒ user-defined routines are routines that users create themselves. User-defined routines provide a means for users to extend the SQL language.

⇒ The user defined routines capture the functionality of most commonly used arithmetic, string, and casting functions. However, these also allow to create routines to encapsulate logic of your own.

⇒ user-defined procedures, functions and methods are created in the database by executing the appropriate CREATE statement for the routine type. These routine creation statements include.

CREATE PROCEDURE
CREATE FUNCTION
CREATE METHOD

FOR example:-

Name of user
Define Routine
↑
CREATE FUNCTION Addition
RETURN NUMBER USING
C NUMBER; (a IN NUMBER, b IN NUMBER)

→ Return value

BEGIN

c := a + b; } Logic of user defined routine
RETURN c;

END

⇒ The clauses specific to each of the CREATE statements define the characteristics of the routine, such as the name, the number and type of routine arguments, and details about the routine logic.

Collection Types:-

⇒ A collection is an ordered group of elements having the same data type.

⇒ The collections are used to store multiple values in a single column of table.

⇒ The collection add flexibility to the database design.

Following are the ways by which the collection types can be used

ARRAY:- It is an entity in that represents one dimensional array with maximum number of elements.

LIST: ordered collection that allows duplicates

SET:- unordered collection that does not allow duplicates.

For example:- Array can be created as follows:-

```

VARCHAR(20) ARRAY[4];
ARRAY['Azechara', 'Ashwini', 'Ashwarya', 'Sharda'];

```

User Defined Types:-

⇒ The user defined type (UDT) is a data type derived from an existing data type.

⇒ UDT allows the database developer to extend the built in types and to create a customized data type.

⇒ There are two types of user defined types:-

⇒ Distinct Type:- A distinct type is a user-defined data type that share its internal representation with an existing built-in data type. For example:-

```
CREATE TYPE empNumberType
AS VARCHAR(5) FINAL;
```

⇒ Structured Type:- A structured type

is

containing one or more named attributes each of which has a data type. Attributes are properties that describe an instance of a type. A structured type also includes a set of method specifications. Methods enable you to define behaviours for structured types. For example:-

```
CREATE TYPE Person AS OBJECT (
  name    VARCHAR2(30),
  Phone    VARCHAR2(20));
```

```
CREATE TYPE Purchase_order AS OBJECT (
  id NUMBER,
```

contact Person;

MEMBER FUNCTION

```
get_value RETURN NUMBER);
```

⇒ We have defined an object named Person and created a structured type Purchase_order.

⇒ The value of the attribute can be accessed using the dot operator.

For example if P is an instance of Person then

P. name = 'Anand'

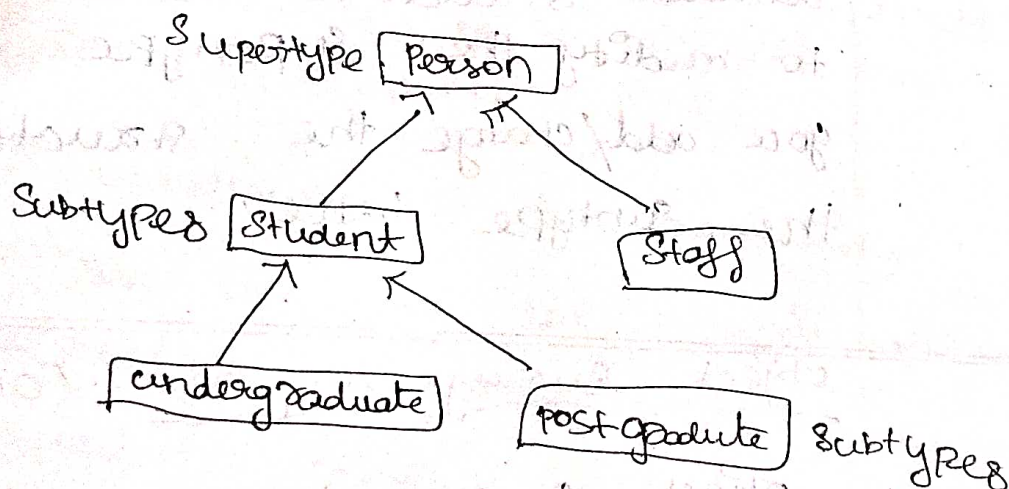
allows to assign name of to the attribute of the Person data type.

Subtypes and Supertypes

Supertype: It an entity type that has got the relationship as parent to child with one or more subtypes. It contains attributes that are common to its subtypes.

Subtype: The subtypes are group of subtype entity and have unique attributes but they are different from each subtype.

For example:-



To create a subtype StudentType of the Supertype: Person type we write


```
CREATE TABLE
AS C
ROUND VARCHAR2,
name VARCHAR(10),
branchID CHAR(4)
```

Advantages of Subtype and Supertype.

=> Avoids data redundancy, as each Subtype contains only unique information. This in turn leads to a reduction in the size of the database itself.

Reduction of errors in programming operations on a table that corresponds to a specific subtype of an entity.

Flexibility in modifying the database structure. There is no need to modify the supertype table if you add/change the structure of the subtype table.

Object Query Language (OQL)

Object Query Language (OQL) is a query language standard for object-oriented databases modeled after SQL.

The following rules apply to OQL statements.

⇒ All complete statements must be terminated by a semi-colon.

⇒ A list of entries in OQL is actually separated by commas but not terminated by a comma (,). Strings of text are enclosed by matching quotation marks.

Basic Form of OQL: SELECT, FROM, WHERE.

SELECT <list of values>

FROM <list of collections and variable assignments>

WHERE <condition>

⇒ Where the SELECT clause extracts those elements of a collection meeting a specific condition.

DOT notations and Path expressions

We use the dot notation and path expressions to access components of complex values.

Example → salary

Example → Students → Set of tuples of type tuple (name: string, fee: real)

Salary $\rightarrow$ real.

CAP Theorem :-

$\Rightarrow$ CAP theorem is also called as Brewer's theorem.

$\Rightarrow$ The CAP theorem is comprised of three components as they relate to distributed data stores:

$\Rightarrow$ Consistency: All reads receive the most recent write or an error.

$\Rightarrow$ Availability: All reads contain data but it might not be the most recent.

$\Rightarrow$ Partition tolerance: The system continues to operate despite network failures (i.e) dropped partitions, slow network connections, or unavailable network connections between nodes)

$\Rightarrow$ The CAP theorem states that it is not possible to guarantee all three of the desirable properties: consistency, availability, and partition tolerance at the same time in a distributed system with data replication.

Document - based: MongoDB Data
and CRUD operation.

⇒ MongoDB is an open source, document based database.

⇒ It is developed and supported by a company named 10gen which is now known as MongoDB Inc.

Why MongoDB is needed?

There are so many efficient RDBMS products available in the market, then why do we need MongoDB? Well, all the modern applications require Big data, faster development and flexible deployment. This need is satisfied by the document based database like MongoDB.

Features of MongoDB.

- 1) It is a schema-less, document based database system.
- 2) It provides high performance data persistence.
- 3) It supports multiple storage engines.

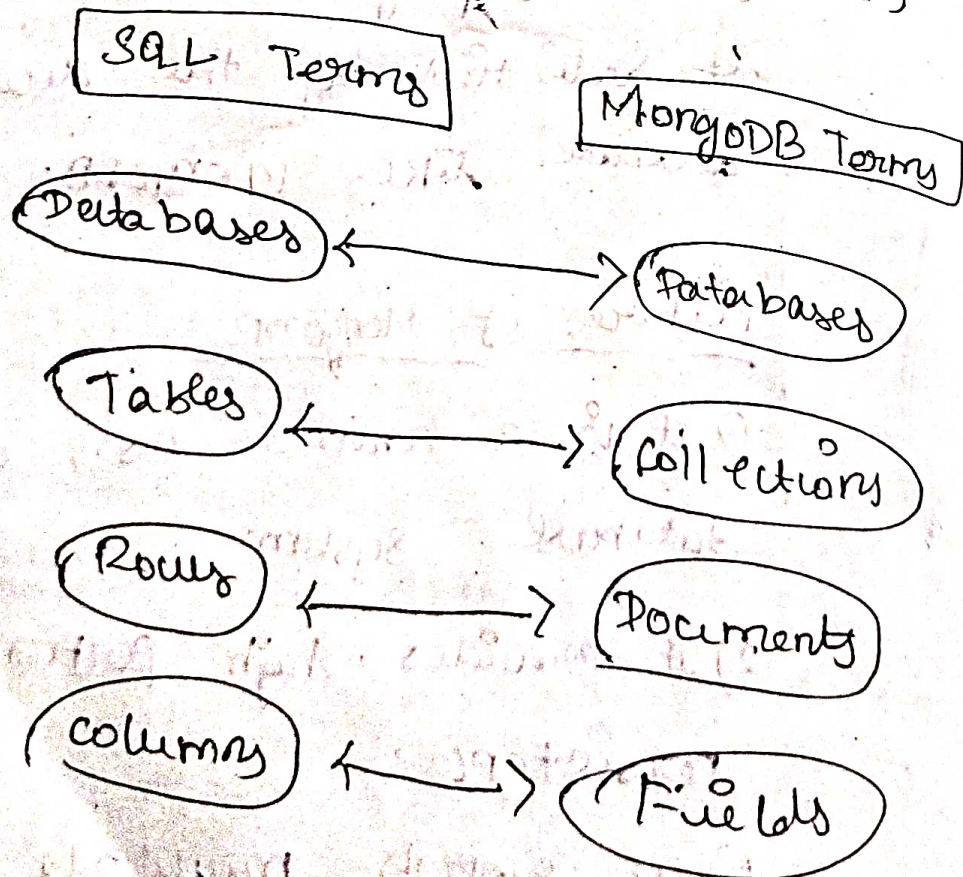
4) It has a rich query language support.

5) MongoDB provides high availability and redundancy with the help of replication.

6) MongoDB provides horizontal scalability with the help of sharding. Sharding means to distribute data on multiple servers.

SQL structure vs. MongoDB.

⇒ In the same manner, the rows in RDBMS are called documents in MongoDB, likewise the columns of the record in RDBMS are called fields.



To the left hand side we show the database in the form of table and to the right hand side the database is shown in the form of collection.

Name	course	Address Line 1	Address Line 2
Ankita	computer engineering	Lane 1	lane 2
Prajita	Mechanical engineering	Street 1	Street 2
Kavita	civil engineering	Road 1	Road 2

Data Types:-

1) Integers - This data type is used for storing the numerical value.

click on Finish button, to complete the installation process.

⇒ In order to verify whether it is installed correctly or not, go to command prompt window and execute the command for magod.exe file's version. It is illustrated as follow.

Microsoft Windows (version 10.0.19041.804)
(c) 2020 Microsoft Corporation. All rights reserved.

C:\Users\DELL> "c:\Program Files\MongoDB\Server\4.4\bin\mongod.exe" --version

Issue this command

```
"version": "4.4.4"
"gitversion": "8db30ab3bb1a9d84"
"modules": [ ],
"allocator": "tcmalloc",
```

And you will get this result if mongod is installed correctly!!

To Set the environment variable, open System properties and click on Environment Variables.

System Properties

Computername

Hardware

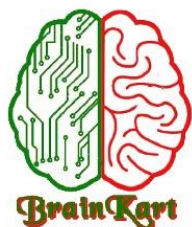
Advanced

System

Remote

Protection

you must be logged on as an Administrator to make most of these changes.



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